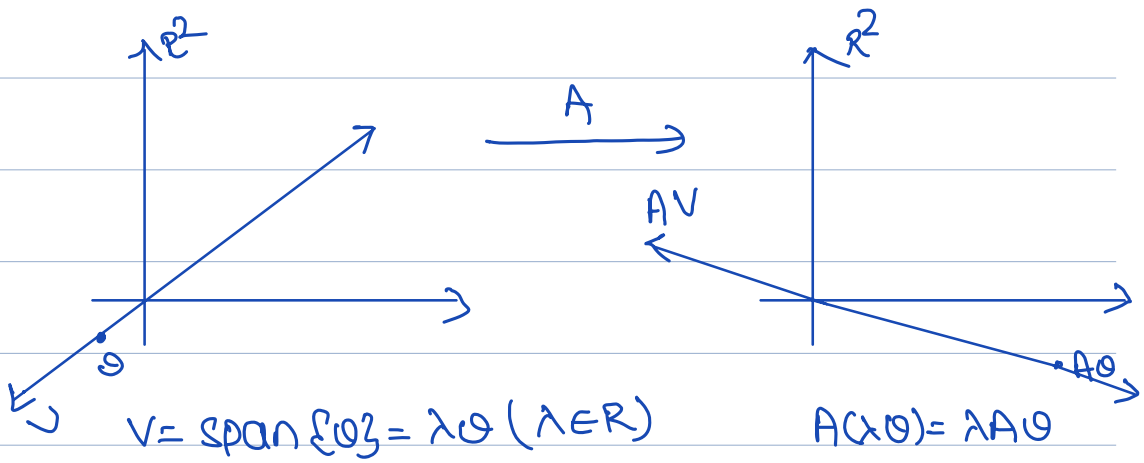
Let $A \in R^{2\times 2}$ be a map from $R^{2\times 1} \to R^{2\times 1}$



This transformation maps straight line through origin to straight lines or points.

We're interested in determining whether there exists a straight line that's mapped to itself under the transformation.

$\Rightarrow A0 = \lambda 0$

$A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ $\quad A\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} y \\ -x \end{bmatrix}$ rotates vector by $\pi/2$ anticlockwise

$\Rightarrow Aw \neq \lambda w \quad \forall w$

$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = A\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$ infinitely many w's

$A \in \mathbb{F}^{n \times n}$. A scalar $\lambda \in \mathbb{F}$ is said to be eigenvalue of A if there exists a non-zero vector $x \in \mathbb{F}^{n \times 1}$ such that $Ax = \lambda x$. x is called eigenvector corresponding to eigenvalue λ.

For a single eigenvalue there are infinitely many eigenvectors (i.e αx, $\alpha \in \mathbb{F} - \{0\}$).

eigenvalue can be 0 but eigenvector has to be non-zero.

$A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \qquad x \neq 0 = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

Let $Ax = \lambda x \Rightarrow \begin{bmatrix} x_2 \\ -x_1 \end{bmatrix} = \begin{bmatrix} \lambda x_1 \\ \lambda x_2 \end{bmatrix} \Rightarrow x_1 = x_2 = 0$

$\downarrow$

no eigenvalue

$A\begin{bmatrix} 1 \\ i \end{bmatrix} = \begin{bmatrix} i \\ -1 \end{bmatrix} = i\begin{bmatrix} 1 \\ i \end{bmatrix} \qquad A\begin{bmatrix} 1 \\ -i \end{bmatrix} = \begin{bmatrix} -i \\ -1 \end{bmatrix} = -i\begin{bmatrix} 1 \\ -i \end{bmatrix}$

$\Rightarrow$ i and $-i$ are complex eigenvalues with associated eigenvectors $\begin{bmatrix} i \\ -1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ -i \end{bmatrix}$ respectively

$Ax = \lambda x$ (x is non-zero)

$\Rightarrow (A - \lambda I)x = 0$ $\rightarrow$ has a non-trivial solution

$\Rightarrow (A - \lambda I)$ is not-invertible

$\Rightarrow |A - \lambda I| = 0$

$\lambda \in F$ is an eigenvalue of $A \in \mathbb{F}^{n \times n}$ iff $|A - \lambda I| = 0$

Eigenspace associated with eigenvalue λ

$$E_\lambda(A) := \{x \in \mathbb{F}^{n \times 1} : Ax = \lambda x\} = Sol(A - \lambda I, 0)$$

$0 \in E_\lambda(A)$. All non-zero elements of $E_\lambda(A)$ are eigenvectors of $A \in \mathbb{F}^{n \times n}$ corresponding to eigenvalue λ.

Characteristic polynomial: $\chi_A(t) = \det(tI - A)$

Maximum number of eigenvalues of $A \in \mathbb{F}^{n \times n}$ is n.

Spectrum: I^n has characteristic polynomial $\chi(t) = (t-1)^n$, and 1 is the only eigenvalue of I. But as a root it is appearing n times.

(including multiplicity)

The multiset of all complex eigenvalues of A is called spectrum of A $\sigma(A)$.

$\sigma(I) = \{1, 1, \ldots 1\}$ (n-times)

$\sigma(A)$ should have n-elements where A is $F^{n\times n}$

$$A = \begin{bmatrix} 2 & i & 0 \\ -i & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

$$\chi_A(t) = |tI - A| = \begin{vmatrix} t-2 & -i & 0 \\ i & t-2 & 0 \\ 0 & 0 & t-3 \end{vmatrix}$$

$$= (t-2)(t-2)(t-3) - 1(t-3) = (t-3)^2(t-1)$$

$$\sigma(A) = \{1, 3, 3\}$$

$\lambda = 1$

$$Ax = x \Rightarrow \begin{aligned} 2x_1 + ix_2 &= x_1 \\ -ix_1 + 2x_2 &= x_2 \\ 3x_3 &= x_3 \Rightarrow x_3 = 0 \end{aligned} \Rightarrow x = \alpha \begin{bmatrix} -i \\ 1 \\ 0 \end{bmatrix}$$

$\alpha \in F - \{0\}$

$\lambda = 3$

$Ax = 3x$ $\quad 2x_1 + ix_2 = 3x_1$

$-ix_1 + 2x_2 = 3x_2$

$3x_3 = 3x_3$

$x = \beta \begin{bmatrix} i \\ 1 \\ 0 \end{bmatrix} + \gamma \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

$\beta, \gamma \in \mathbb{F} - \{0\}$

- $\sigma(A) = \sigma(A^t)$
- If $B \in \mathbb{F}^{n \times n}$ is similar to A; $\sigma(A) = \sigma(B)$
- If A is triangular matrix, then $\sigma(A)$ is the diagonals of A

Eigenvectors corresponding to distinct eigenvalues of a square matrix are linearly independent

Let $A \in \mathbb{F}^{n \times n}$ and $\sigma(A) = \{\lambda_1, \lambda_2, \ldots \lambda_n\}$. Then

(i) $\det A = \lambda_1 \lambda_2 \ldots \lambda_n$

(ii) $tr(A) = \lambda_1 + \lambda_2 + \cdots + \lambda_n$

Special types of matrices

Symmetric: $A^t = A$ $\quad$ Skew Symmetric: $A^t = -A$

Hermitian: $A^* = A$ $\quad$ Skew hermitian: $A^* = -A$

unitary: $A^*A = I = AA^*$ $\quad$ orthogonal: $A^tA = I = AA^t$

Isometry: $A \in \mathbb{F}^{m\times n} \Rightarrow A^*A = I_n$ and $AA^* = I_m$

$A = \frac{1}{2}[A + A^t] + \frac{1}{2}[A - A^t]$ = symmetric + skewsym

$A = \frac{1}{2}[A + A^*] + \frac{1}{2}[A - A^*]$ = hermitian + skew herm

$A = \frac{1}{2}[A + A^*] + i\,\frac{1}{2i}[A - A^*]$ = hermitian + i.hermitian

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$x = (x_1, \dots x_n) \in \mathbb{F}^n$, $y = (y_1, \dots, y_n) \in \mathbb{F}^n$

$\langle x\ y \rangle = \sum_{i=1}^{n} x_i \overline{y_i}$

let $A \in \mathbb{C}^{n\times n} \Rightarrow \langle Ax\ y \rangle = \langle x\ A^*y \rangle$

Let λ be eigenvalue of $A \in \mathbb{F}^{n\times n}$

- If A is hermitian, then $\lambda \in \mathbb{R}$
- If A is skewhermitian, then λ is either zero or purely complex
- If A is unitary, then $|\lambda| = 1$ and $\det(A) = 1$

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① A is unitary ② $A^*A = I$ ③ $AA^* = I$

④ Columns of A form an orthormal basis

⑤ Rows of A form an orthormal basis

⑥ $\langle Ax\ Ay\rangle = \langle x\ y\rangle \quad \forall\ x,y \in C^n$

⑦ $\|Ax\| = \|x\| \quad \forall\ x \in C^n$

The above 7 points are equivalent for $A \in C^{n\times n}$

proof: clearly ① ⇒ ② ① ⇒ ③

$$A = [c_1 \ldots c_n] \qquad A^* = [c_1^* \ c_2^* \ldots c_n^*]^t$$

($c_1, \ldots, c_n$: columns)

$$A^*A = \begin{bmatrix} c_1^*c_1 & c_1^*c_2 & \ldots & c_1^*c_n \\ c_2^*c_1 & c_2^*c_2 & \ldots & c_2^*c_n \\ \vdots & \vdots & & \vdots \\ c_n^*c_1 & c_n^*c_2 & \ldots & c_n^*c_n \end{bmatrix} = [\langle c_j\ c_i\rangle]$$

$$A^*A = I \iff \begin{cases} c_p^* c_q = 0 & \text{for } p \neq q \\ c_p^* c_q = 1 & \text{for } p = q \end{cases} \Big\} \text{ orthonormal basis } ④$$

② ⇒ ②⇔④

Similarly we can say ③⇔⑤ by taking rows of A.

$$A^*A = I \Rightarrow \det(A^*A) = 1 = \det(A^*)\det(A) = 1$$

② ⇒ A^*, A are invertible

$AA^* = A(A^*A)A^* = (AA^*)^2 \Rightarrow AA^* = I$ (Since AA^* is invertible)

③

$\Rightarrow$ 2 $\Leftrightarrow$ ③

① $\Rightarrow$ ⑥ is trivial

$\langle Ax, Ay \rangle = \langle x\ A^*Ay \rangle = \langle x\ y \rangle$

⑥ $\Rightarrow$ ⑦ is trivial

$\langle Ax, Ax \rangle = \langle x\ A^*Ax \rangle = \langle x, x \rangle$

$\Rightarrow \|Ax\| = \|x\|$

Let $A \in C^{n \times n}$ be hermitian, then the eigenvectors corresponding to distinct eigenvalues are orthogonal.

proof $\lambda, \mu \rightarrow$ 2 distinct eigenvalues (real)

$\rightarrow (\lambda - \mu)\langle x, y \rangle = \lambda \langle x\ y \rangle - \mu \langle x\ y \rangle$

$= \langle \lambda x, y \rangle - \langle x, \mu y \rangle$

$= \langle Ax, y \rangle - \langle A^*x, y \rangle$

$= 0$

$\Rightarrow \langle x, y \rangle = 0$

If $T: F^{n\times 1} \to F^{n\times 1} \Rightarrow \exists\ A \in F^{n\times n} \Rightarrow Tx = Ax$

$$A = \left[[T(e_1)]_B, \ldots, [T(e_n)]_B \right] \quad A_{B,B}$$

If B' is a non-standard basis for F^n

$$[f_1\ f_2, \ldots f_n]$$

$$\Rightarrow \quad P^{-1}\underbrace{\left[[T(f_1)]_{B'}, \ldots, [T(f_n)]_{B'} \right]}_{T_{B',B'}} P = A$$

we want to find a basis B such that $[T]_{B',B'}$ is a diagonal matrix

Schor's triangularization Theorm:

Let $A \in C^{n\times n}$. Then there exists (need not be unique) a unitary matrix $P \in C^{n\times n}$ such that $P^{-1}AP$ or P^*AP (both are same) is upper triangular. Furthermore if A is a real matrix and all eigenvalues are real, the P can be chosen to be a orthogonal matrix

proof To prove $\exists$ a unitary $P \Rightarrow P^*AP$ is upper triangular

$P^*AP = B \Rightarrow A = PBP^*$ (A and B are similar)

$\Rightarrow AP = PB$ $\Rightarrow$ Their spectrum is same)

$\Rightarrow \sigma(A) = \sigma(B) = \text{diag}(B)$

If $P = [c_1\ c_2 \dots c_n] \Rightarrow AP = [Ac_1\ Ac_2 \dots Ac_n]$

$PB = [\lambda_1 c_1 \dots]$ $\quad B = \begin{bmatrix} \lambda_1 & \cdots \\ 0 & \vdots \end{bmatrix}$

Column 1 has to be same atleast

$Ac_1 = \lambda_1 c_1 \qquad c_1 \neq 0$

$\Rightarrow c_1$ is an eigenvector of A corresponding to eigenvalue λ_1

How to set P and B ?

- Fix one eigenvalue of A say λ_1 and a corresponding eigenvector say $u_1 \in \mathbb{C}^n$ with $\|u_1\| = 1$
- Now extend this to an orthonormal basis say $u_1, u_2, \dots u_n$.
- Now let $P_1 = [u_1\ u_2 \dots u_n] \in \mathbb{C}^{n \times n}$ is unitary
- $P_1^* A P_1 = \left[\begin{array}{c|c} \lambda_1 & \cdots \\ \hline 0 & \\ 0 & A_1 \\ 0 & \end{array}\right]$

- $A_1 \in \mathbb{C}^{(n-1)\times(n-1)}$. Apply the same steps and find $P_2 \in \mathbb{C}^{(n-1)\times(n-1)}$ unitary such that

$$P_2^* A_1 P_2 = \left[\begin{array}{c|c} \lambda_2 & \cdots \\ \hline 0 \\ 0 \\ \vdots & A_2 \end{array}\right]$$

$$\Rightarrow A = P_1^* \left[\begin{array}{c|c} \lambda_1 & \cdots \\ \hline 0 & \\ 0 & P_2 \left[\begin{array}{c|c} \lambda_2 & \cdots \\ \hline 0 \\ 0 \\ \vdots & A_2 \end{array}\right] P_2^* \\ 0 & \\ 0 & \end{array}\right] P_1$$

$$= P_1^* \left[\begin{array}{c|c} 1 & \\ \hline 0 & \\ 0 & P_2 \\ 0 & \end{array}\right] \left[\begin{array}{cc|c} \lambda_1 & \cdots & \cdots \\ 0 & \lambda_2 & \cdots \\ \hline 0 & 0 & A_2 \\ 0 & 0 & \end{array}\right] \left[\begin{array}{c|c} 1 & \\ \hline & P_2^* \end{array}\right] P_1$$

example: $A = \begin{bmatrix} 2 & 1 & 0 \\ 2 & 3 & 0 \\ -1 & -1 & 1 \end{bmatrix}$: $\sigma(A) = \{1, 1, 4\}$

$\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ is an eigenvector corresponding to eigenvalue 1 and $\|u\| = 1$ $\lambda_1 = 1$

Notice that $\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ is an ONB.

let $P_1 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$

$$\Rightarrow P_1^* A P_1 = \begin{bmatrix} 1 & -1 & -1 \\ 0 & 2 & 1 \\ 0 & 2 & 3 \end{bmatrix}$$

A_1

eigen value of A_1: 1, 4

$\begin{bmatrix} 1 \\ -1 \end{bmatrix}$ is an eigenvector corresponding to $\lambda = 1$

$$\Rightarrow P_2 = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

$$P_2^* A_1 P_2 = \begin{bmatrix} 1 & -1 \\ 0 & 4 \end{bmatrix}$$

$$\Rightarrow P = P_1 \begin{bmatrix} 1 & 0 \\ 0 & P_2 \end{bmatrix} = \begin{bmatrix} 0 & 1/\sqrt{2} & 1/\sqrt{2} \\ 0 & -1/\sqrt{2} & 1/\sqrt{2} \\ 1 & 0 & 0 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 0 & -\sqrt{2} \\ 0 & 1 & -1 \\ 0 & 0 & 4 \end{bmatrix} = P^* A P$$

Annihilating polynomials:

Given a polynomial $p(t) = c_0 + c_1 t + \dots + c_k t^k$

we define $p(A) = c_0 I + c_1 A + c_2 A^2 + \dots + c_k A^k$

A polynomial $p(t)$ is said to be annhilates the matrix $A \in \mathbb{F}^{n\times n}$ if $p(A) = 0^n$

Cayley Hamilton: If $A \in \mathbb{F}^{n\times n}$, then its characteristic polynomial annhilates A i.e

$\chi_A(A) = 0$

$\chi_A(A) = c_0 I + c_1 A + c_2 A^2 + \cdots + c_n A^n$

$\chi_A(A) = 0 \Rightarrow c_n A^n = -(c_0 I + c_1 A + \cdots + c_{n-1} A^{n-1})$

$\chi_A(t) = (t-\lambda_1)(t-\lambda_2)\cdots(t-\lambda_n) = (-1)^n \lambda_1\lambda_2\cdots\lambda_n + \cdots\cdots t^n$

$\Rightarrow c_0 = (-1)^n \det(A)$

$A^{-1} = \frac{-1}{c_0}\left[c_1 I + c_2 A + \cdots + c_n A^{n-1}\right]$

Minimal polynomial:

Let $A \in C^{n\times n}$, then

(i) there exists a unique monic polynomial of least degree that annhilates A.

(ii) This unique polynomial is called minimal polynomial of A, denoted by $m_A(t)$

(iii) The minimal polynomial divides the characteristic polynomial : $\chi_A(t) = m_A(t) \cdot p(t)$ for some polynomial $p(t)$

(iv) Both $\chi_A(t)$ and $m_A(t)$ have same set of zeros (possibly different multiplicity)

Self observation: If A is a diagonal matrix and $\chi_A(t) = (t-\lambda_1)^{m_1}(t-\lambda_2)^{m_2}\ldots$ Then $m_A(t) = (t-\lambda_1)(t-\lambda_2)\ldots$

Diagonalizable matrix:

A square matrix A is said to be diagonalizable if there exist an invertible $P \in F^{n\times n}$ such that $P^{-1}AP$ is a diagonal matrix.

→ $D = P^{-1}AP$ → diagonal $\Rightarrow A = PDP^{-1}$

A is similar to a diagonal matrix

→ All diagonal matrices are diagonalizable

→ If A is diagonizable, then A^k is also

diagonizable ($k \in \mathbb{N}$)

$\rightarrow \sigma(A) = \sigma(D) = \{d_1, d_2, \dots d_n\}$

let $P = [u_1\ u_2 \dots u_n]$ $\quad \{u_i \neq 0\ \forall\ i \in [1,n]\}$

$\Rightarrow AP = PD$

$\Rightarrow [Au_1\ Au_2 \dots Au_n] = [d_1u_1\ d_2u_2 \dots d_nu_n]$

$\Rightarrow Au_i = d_iu_i \quad \forall \quad 1 \leq i \leq n$

$\Rightarrow u_i$ is an eigenvector of A with eigenvalue d_i

The following are equivalent for $A \in \mathbb{F}^{n \times n}$

① A is diagonalizable

② There exists a basis consisting of eigenvectors of A

③ The geometric multiplicity of each eigenvalue of A is equal to its geometric multiplicity.

④ Sum of geometric multiplicities of all eigenvalues of A is n

⑤ There exists distinct scalars $\lambda_1, \lambda_2, \dots \lambda_k$ such that $m_A(t) = (t-\lambda_1)(t-\lambda_2)\dots(t-\lambda_k)$

Singular Value Decomposition:

A matrix $B \in C^{n \times n}$ is said to be positive semidefinite if $\langle Bx, x \rangle \geq 0 \quad \forall \quad x \in C^{n \times 1}$

Let $B \in C^{n \times n}$, then the following are equivalent:

(a) B is positive (semidefinite)

(b) B is normal and $\sigma(B) \subseteq [0, \infty)$

(c) B is self adjoint and $\sigma(B) \subseteq [0, \infty)$

(d) $B = A^* A$ for some $A \in C^{m \times n}$ for some $m \in N$

$\Rightarrow \sigma(A^* A) \subseteq [0, \infty)$ from above theorm

$\sigma(AA^*) \subseteq [0, \infty)$

Singular values: Let $A \in C^{m \times n}$ and let $\sigma(A^* A) = \{\lambda_1, \lambda_2, \ldots, \lambda_n\}$. Then non-negative square roots $s_i := \sqrt{\lambda_i}$; $1 \leq i \leq n$, are called singular values of A.

Let $A \in C^{m \times n}$. Then $\text{rank}(A) = \text{rank}(A^*A)$

$= \text{rank}(AA^*) = \text{rank}(A^*) =$ number of positive singular values of A.

A positive real number is an eigenvalue of A^*A iff it is an eigenvalue of AA^*

Actual theorm: SVD

Let $A \in C^{m \times n}$ be of rank r. Let $s_1, s_2, \ldots, s_r$ be positive singular values of A, and

$$\Sigma = \begin{bmatrix} S & 0_{m \times (n-r)} \\ 0_{(m-r) \times n} & 0_{(m-r) \times (n-r)} \end{bmatrix} \in C^{m \times n}$$

$S = \text{diag}(s_1, s_2, \ldots s_r) \in C^{r \times r}$

preferably $s_1 > s_2 > s_3 > \ldots s$

There exists unitary matrices

$U = [u_1, \ldots u_m] \in C^{m \times m}$ and $V = [v_1 \ldots v_n] \in C^{n \times n}$,

the following holds: u_i's → left singular vector
v_i's → right singular vectors

1. $A = U\Sigma V^* = s_1 u_1 v_1^* + s_2 u_2 v_2^* + \cdots + s_r u_r v_r^*$

s_i, u_i, v_i → column vectors

2. $AA^* u_i = \begin{cases} s_i^2 u_i & 1 \le i \le r \\ 0 & r < i \le m \end{cases}$

$A^* A v_i = \begin{cases} s_i^2 v_j & 1 \le j \le r \\ 0 & r < j \le n \end{cases}$

3. $u_i = (s_i)^{-1} A v_i$ and $v_i = (s_i)^{-1} A^* u_i \quad \forall\ 1 \le i \le r$ $\quad (s_i)^{-1} = 1/s_i$

4. (a) $\{u_1, u_2, \ldots u_r\}$ is a ONB for $R(A)$

(b) $\{u_{r+1}, u_{r+2}, \ldots u_m\}$ is a ONB for $N(A^*)$

(c) $\{v_1, v_2, \ldots v_r\}$ is a ONB for $R(A^*)$

(d) $\{v_{r+1}, v_{r+2}, \ldots v_n\}$ is a ONB for $N(A)$

Find SVD of $A = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}_{2\times 4}$

$AA^* = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \Rightarrow \sigma(AA^*) = \{2, 2\}$

$\Rightarrow s_1 = \sqrt{2},\ s_2 = \sqrt{2}$

$\Sigma = \begin{bmatrix} \sqrt{2} & 0 & 0 & 0 \\ 0 & \sqrt{2} & 0 & 0 \end{bmatrix}$

$$N(AA^* - s_i^2 I) = N(0) = \mathbb{C}^2 = \text{span}(e_1, e_2)$$

$$\Rightarrow u_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad u_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad U = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$v_i = \frac{Au_i}{s_i} \quad (i = 1,2) \Rightarrow v_1 = \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \\ 0 \end{bmatrix} \quad v_2 = \begin{bmatrix} 0 \\ 1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix}$$

$$v_3, v_4 \Rightarrow N(A^*A) = N(A) = \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ -1/\sqrt{2} \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1/\sqrt{2} \\ 0 \\ -1/\sqrt{2} \end{bmatrix}$$

$$V = \begin{bmatrix} 1/\sqrt{2} & 0 & 1/\sqrt{2} & 0 \\ 0 & 1/\sqrt{2} & 0 & 1/\sqrt{2} \\ 1/\sqrt{2} & 0 & -1/\sqrt{2} & 0 \\ 0 & 1/\sqrt{2} & 0 & -1/\sqrt{2} \end{bmatrix}$$

$$A = U\Sigma V^*$$

Thin SVD:

$A = U\Sigma V^*$

Let $m \leq n$

$$\Sigma = \begin{bmatrix} S & 0_{r\times(n-r)} \\ 0_{(m-r)\times r} & 0_{(m-r)\times(n-r)} \end{bmatrix}$$

$$= \begin{bmatrix} \Sigma_1 & 0_{m\times(n-m)} \end{bmatrix}$$

where $\Sigma_1 = \text{diag}(s_1, s_2 \dots s_r, 0, 0, \dots 0) \in C^{m\times m}$

let $u_1 = u$

let $V = [v_1\ v_2 \dots v_m\ v_{m+1}\ v_{m+2} \dots v_n] = [V_1\ V_2] \in C^{n\times n}$

where columns of V_1 are orthonormal

$$\Rightarrow u\Sigma V^* = u_1 [\Sigma_1\ 0] \begin{bmatrix} V_1^* \\ V_2^* \end{bmatrix} = u_1 \Sigma_1 V_1^*$$

$u_1 \in C^{m\times m}$ is unitary

$V_1 \in C^{n\times m}$ is having orthonormal columns

$\Sigma_1 \in C^{m\times m}$

Similarly is $m \geq n$:

$$\Sigma_2 = \text{diag}(s_1, s_2, \dots s_r, \dots 0, 0) \in C^{n\times n}$$

$$\Rightarrow \Sigma = \begin{bmatrix} \Sigma_2 & 0_{(m-n)\times n} \end{bmatrix}^t$$

let $V_2 = V$

let $u_2 = [u_1\ u_2 \dots u_n\ u_{n+1} \dots u_m] = [u_2\ u_3] \in C^{m\times m}$

where columns of U_2 are orthonormal.

$$\Rightarrow U\Sigma V^* = [U_2\ U_3]\begin{bmatrix}\Sigma_2\\ 0\end{bmatrix} V_2^* = U_2\Sigma_2 V_2^*$$

where $V_2 \in C^{n\times n}$ is unitary

$U_2 \in C^{m\times n}$ have orthonormal columns

$\Sigma_2 \in C^{n\times n}$

Summary: If $A \in C^{m\times n}$ has rank r and $k = \min(m,n)$, then $A = \bar{U}\bar{\Sigma}\bar{V}^*$

where $\bar{U} \in C^{m\times k}$; $\bar{\Sigma} = \text{diag}(s_1, s_2 .. s_r, 0, 0..) \in C^{k\times k}$

and $\bar{V} \in C^{n\times k}$; and $\bar{U}, \bar{V}$ have orthonormal columns

Tight SVD:

$$U = [u_1\ u_2 ..\ u_r\ u_{r+1} \cdots u_m] = [\hat{u}\ u_0] \in C^{m\times m}$$

$$V = [v_1\ v_2 ..\ v_r\ v_{r+1} \cdots v_n] = [\hat{V}\ v_0] \in C^{n\times n}$$

$$\Sigma = \begin{bmatrix} S & 0\\ 0 & 0\end{bmatrix} \in C^{m\times n}$$

$$S = \text{diag}(s_1, \cdots s_r) \in C^{r\times r}$$

$\Rightarrow U\Sigma V^* = \hat{U}S\hat{V}^*$

$\hat{U} \in C^{m\times r}$, $\hat{V} \in C^{r\times n}$ $\rightarrow$ have orthonormal columns

let $B = \hat{U}S \in C^{m\times r}$; $C = S\hat{V}^* \in C^{r\times n}$ (both have rank r)

$A = B\hat{V}^* = \hat{U}.C$

is a product of matrices of rank r.

$\Rightarrow$ Above is the full Rank Factorization of A.

$$A = \begin{bmatrix} 2 & -1 \\ -2 & 1 \\ 4 & -2 \end{bmatrix}$$

rank = 1 = no of +ve singular values

$$A^*A = \begin{bmatrix} 24 & -12 \\ -12 & 6 \end{bmatrix} \Rightarrow \sigma(A^*A) = \{0, 30\}$$

$\Rightarrow S_1 = \sqrt{30}$

$$\Rightarrow \Sigma = \begin{bmatrix} \sqrt{30} & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}_{3\times 2} \qquad S = [\sqrt{30}]_{1\times 1}$$

θ_1 must be eigenvector corresponding to A^*A

$$N(A^*A - 30I) = (A^*A)\begin{bmatrix} a \\ b \end{bmatrix} = 30\begin{bmatrix} a \\ b \end{bmatrix}$$

$$\Rightarrow v_1 = \begin{bmatrix} -2/\sqrt{5} \\ 1/\sqrt{5} \end{bmatrix}$$

$$N(A^*A) = (A^*A)\begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \qquad V = \begin{bmatrix} -2/\sqrt{5} & 1/\sqrt{5} \\ 1/\sqrt{5} & 2/\sqrt{5} \end{bmatrix}$$

$$\Rightarrow v_2 = \begin{bmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix}$$

$$u_1 = \frac{1}{\sqrt{30}} \begin{bmatrix} 2 & -1 \\ -2 & 1 \\ 4 & -2 \end{bmatrix} \begin{bmatrix} -2/\sqrt{5} \\ 1/\sqrt{5} \end{bmatrix} = \begin{bmatrix} -1/\sqrt{6} \\ 1/\sqrt{6} \\ -2/\sqrt{6} \end{bmatrix}$$

$$N(AA^*) = (AA^*).u_i = 0$$

$$= \text{span}\left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \right\}$$

$$u_2 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix} \qquad u_3 = \begin{bmatrix} 1/\sqrt{3} \\ -1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix} \Rightarrow U = \begin{bmatrix} -1/\sqrt{6} & 1/\sqrt{2} & 1/\sqrt{3} \\ 1/\sqrt{6} & 1/\sqrt{2} & -1/\sqrt{3} \\ -2/\sqrt{6} & 0 & 2/\sqrt{3} \end{bmatrix}$$

$$A = U\Sigma V^*$$

$$\hat{U} = \begin{bmatrix} -1/\sqrt{6} \\ 1/\sqrt{6} \\ -2/\sqrt{6} \end{bmatrix} \qquad \hat{V} = \begin{bmatrix} -2/\sqrt{5} \\ 1/\sqrt{5} \end{bmatrix} \qquad S = [\sqrt{30}]$$

$$\Rightarrow A = \hat{U}S\hat{V}^*$$

Jordan form:

Jordan block: $J_n(\lambda) =$

$$\begin{bmatrix} \lambda & 1 & & & \\ & \lambda & 1 & & \\ & & \ddots & \ddots & \\ & & & \ddots & 1 \\ & & & & \lambda \end{bmatrix}_{n \times n}$$

(diagonal $= \lambda$
superdiagonal $= 1$
unspecified entries are 0)

A matrix is said to be in Jordan form if

$$A = \begin{bmatrix} J_{n_1}(\lambda_1) & & & \\ & J_{n_2}(\lambda_2) & & \\ & & \ddots & \\ & & & J_{n_K}(\lambda_K) \end{bmatrix}$$

$$\begin{bmatrix} 2 & & \\ & 2 & \\ & & 4 \end{bmatrix} = \text{diag}(J_1(2), J_1(2), J_1(4))$$

$$\begin{bmatrix} 2 & 1 & \\ & 2 & \\ & & 3 \end{bmatrix} = \text{diag}(J_2(2), J_1(3))$$

Each matrix $A \in \mathbb{C}^{n \times n}$ is similar to a matrix in Jordan form

$$A = PJP^{-1} \quad ; \quad P \in \mathbb{C}^{n \times n}$$

Moreover, if $m_k(\lambda)$ is the number of Jordan blocks of order k with diagonal entry λ in J, then

$$m_k(\lambda) = \text{rank}((A-\lambda I)^{k-1}) - 2\,\text{rank}((A-\lambda I)^k) + \text{rank}((A-\lambda I)^{k+1}).$$

The jordan form J of A is unique upto a permutation of diagonal Jordan blocks.

Diagonal entries of J are eigenvalues of A.

Number of Jordan block with diagonal entries $\lambda \in \sigma(A)$ is equal to $GM(\lambda) = \dim(N(A-\lambda I))$

By rank-nullity theorm;

$$m_k(\lambda) = \text{null}((A-\lambda I)^k) - \text{null}((A-\lambda I)^{k-1} - \text{null}(A-\lambda I)^{k+1}$$

Let $J = J_A(\lambda)$; $P = [\theta_1, \theta_2 \ldots \theta_n]$

$AP = JP \Rightarrow [A\theta_1, A\theta_2 \; A\theta_n] = [\lambda\theta_1, \theta + \lambda\theta_2 \quad \theta_{n-1} + \lambda\theta_n]$

$\Rightarrow \theta_1 \in N(A-\lambda I) \quad \theta_2 \in N(A-\lambda I)^2 \quad \theta_n \in N(A-\lambda I)^n$

$$A = \begin{bmatrix} 5 & 4 & 2 & 1 \\ 0 & 1 & -1 & -1 \\ -1 & -1 & 3 & 0 \\ 1 & 1 & -1 & 2 \end{bmatrix} = PJP^{-1}$$

$$\chi_A(t) = (t-1)(t-2)(t-4)^2 \Rightarrow \sigma(A) = \{1,2,4,4\}$$

$$N(A-I) = \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} \qquad GM(1) = 1$$

$$N(A-2I) = [1,-1,0,1]^t \qquad GM(2) = 1$$

$$N(A-4I) = [1,0,-1,1] \qquad GM(4) = 1$$

$$J = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 4 & 1 \\ 0 & 0 & 1 & 4 \end{bmatrix}$$